

The Beacon Theorem (Optional Questions' Answers)

1. A 5-layer square beacon pyramid is built having 2 beacons placed out of a maximum of 16 blocks². Determine the blocks needed to build the beacon including and excluding the beacons.

Givens:

$$m = 5; n_b = 2 \text{ beacons}; A_s = 16 \text{ blocks}^2.$$

Required:

β , b , and σ_l .

Analysis:

To find the deducted beacons, use this equation:

$$\beta = A_s - n_b$$

Then use the second beacon formula and the sum of layers excluding the beacons.

$$b = \sum_{h=0}^m (2h + s)^2 - \beta$$

$$\sigma_l = b - n_b$$

Solution:

$$\beta = A_s - n_b$$

$$\beta = 16 - 2$$

$$\beta = 14 \text{ beacons.}$$

Find s .

$$A_s = s^2$$

$$s = \sqrt{A_s}$$

$$s = \sqrt{16 \text{ blocks}^2}$$

$$s = \sqrt{4^2 \text{ blocks}}$$

$$s = 4 \text{ blocks}$$

Find the total blocks needed including the number of beacons.

$$b = \sum_{h=0}^m (2h + s)^2 - \beta$$

$$b = \sum_{h=0}^5 (2h + 4)^2 - 14$$

$$b = ((2(0) + 4)^2 + (2(1) + 4)^2 + (2(2) + 4)^2 + (2(3) + 4)^2 + (2(4) + 4)^2 + (2(5) + 4)^2) - 14$$

$$b = ((4)^2 + (2 + 4)^2 + (4 + 4)^2 + (6 + 4)^2 + (8 + 4)^2 + (10 + 4)^2) - 14$$

$$b = (4^2 + 6^2 + 8^2 + 10^2 + 12^2 + 14^2) - 14$$

$$b = (16 + 36 + 64 + 100 + 144 + 196) - 14$$

$$b = 556 - 14$$

$$b = 542 \text{ blocks}^2$$

Find the total blocks needed excluding the number of beacons.

$$\sigma_l = b - n_b$$

$$\sigma_l = 542 - 2$$

$$\sigma_l = 540 \text{ blocks}^2$$

Statement:

The total blocks needed to build a 5-layer square beacon pyramid with 2 beacons placed is 542 blocks². Excluding the beacons, the total blocks will be 540 blocks².

2. A beacon has its 3rd layer to be 91 blocks². Determine the length and the width of the maximum beacons and find the area of the total maximum beacons.

Givens:

$$h = 3; b_l = 91 \text{ blocks}^2.$$

Required:

L , W , and A_s .

Analysis:

$$b_l = (2h + L)(2h + W)$$

$$b_l = L_B W_B$$

Solution:

To find L and W , L_B and W_B must be determined.

$$b_l = L_B W_B$$

$$91 = L_B W_B$$

Remember that the first beacon formulas assumed the smallest possible areas. This means that in any type of beacon pyramid, the sides of the beacon layers should not be lower than the minimum.

Mathematically, at h , $S_B \geq (2h + 1)$. Since $h = 3$, $S_B \geq 7$ blocks.

Similarly, $L_B \geq 7$ blocks and $W_B \geq 7$ blocks.

To find L_B and W_B , factor 91 such that both integers must be 7 blocks or more.

$$91 = 91 \times 1 \text{ and } 91 = 13 \times 7$$

$91 = 91 \times 1$ is refused because $W_B \geq 7$ blocks and 1 is less than 7.

$91 = 13 \times 7$ is accepted since 13 blocks ≥ 7 blocks, and 7 blocks ≥ 7 blocks.

Label the variables.

$$91 = L_B W_B$$

$$91 = 13 \times 7$$

$$L_B = 13 \text{ blocks}$$

$$W_B = 7 \text{ blocks}$$

Find L and W .

$$L_B = 2h + L; W_B = 2h + W$$

$$L = L_B - 2h; W = W_B - 2h$$

$$L = 13 - 2(3); W = 7 - 2(3)$$

$$L = 13 - 6; W = 7 - 6$$

$$L = 7 \text{ blocks}$$

$$W = 1 \text{ block}$$

Find A_s .

$$A_s = LW$$

$$A_s = 7(1)$$

$$A_s = 7 \text{ blocks}^2$$

Statement:

The length of the maximum beacons is 7 blocks while the width of the maximum beacons is 1 block. This makes the area of the total maximum beacons is 7 blocks².

3. If $A_s = 5 \text{ blocks}^2$, find the total blocks needed to build a 7-layer beacon pyramid including the beacons. Assume that $\beta = 0$.

Givens:

$$m = 7; A_s = 5 \text{ blocks}^2; \beta = 0 \text{ beacons.}$$

Required:

L , W , and b .

Analysis:

$$A_s = LW$$

$$b = \sum_{h=0}^m (2h + L)(2h + W) - \beta$$

Solution:

To find L and W , factor out A_s such that $s \geq 1$ block. Meaning that $L \geq 1$ block and $W \geq 1$ block.

$$A_s = LW$$

$$5 = LW$$

Since 5 is a prime number, the common factors of 5 are only 5 and 1.

$$5 = 5 \times 1$$

$$L = 5 \text{ blocks}$$

$$W = 1 \text{ block}$$

Find b assuming that $\beta = 0$ beacons.

$$b = \sum_{h=0}^m (2h + L)(2h + W) - 0$$

$$b = \sum_{h=0}^m (2h + L)(2h + W)$$

Substitute the givens.

$$b = \sum_{h=0}^7 (2h + 5)(2h + 1)$$

$$b = \sum_{h=0}^7 4h^2 + 10h + 2h + 5$$

$$b = \sum_{h=0}^7 4h^2 + 12h + 5$$

$$\begin{aligned} b = & (4(0)^2 + 12(0) + 5) + (4(1)^2 + 12(1) + 5) \\ & + (4(2)^2 + 12(2) + 5) + (4(3)^2 + 12(3) + 5) \\ & + (4(4)^2 + 12(4) + 5) + (4(5)^2 + 12(5) + 5) \\ & + (4(6)^2 + 12(6) + 5) + (4(7)^2 + 12(7) + 5) \end{aligned}$$

$$\begin{aligned} b = & (5) + (4 + 12 + 5) + (16 + 24 + 5) + (36 + 36 + 5) \\ & + (64 + 48 + 5) + (100 + 60 + 5) + (144 + 72 + 5) \\ & + (196 + 84 + 5) \end{aligned}$$

$$b = 5 + 21 + 45 + 77 + 117 + 165 + 221 + 285$$

$$b = 936 \text{ blocks}^2$$

Statement:

The total blocks needed to build a 7-layer beacon pyramid with a 5 blocks² area of the total maximum beacons is 936 blocks².

4. If a beacon pyramid with $A_s = 1$ blocks², determine the total blocks needed in layer 21, and calculate the total blocks needed to build a

beacon pyramid with a height of 22 blocks including and excluding the beacon(s).

Givens:

$$h = 21; m = 21; A_s = 1 \text{ blocks}^2.$$

Required:

b_l , b , and σ_l .

Analysis:

Since $A_s = 1 \text{ blocks}^2$, there are no deducted beacons because it is the only beacon, and since 1 is a square number, this makes the beacon pyramid a square beacon pyramid. Hence, the first beacon layer formula, the first beacon formula, and the sum of layers excluding the beacons will be used.

$$b_l = (2h + 1)^2$$

$$b = \sum_{h=0}^m (2h + 1)^2$$

$$\sigma_l = b - 1$$

Solution:

Find b_l .

$$b_l = (2h + 1)^2$$

$$b_l = (2(21) + 1)^2$$

$$b_l = (42 + 1)^2$$

$$b_l = (43)^2$$

$$b_l = 1849 \text{ blocks}^2$$

Find b .

$$b = \sum_{h=0}^m (2h+1)^2$$

$$b = \sum_{h=0}^{21} (2h+1)^2$$

$$\begin{aligned} b = & (2(0) + 1)^2 + (2(1) + 1)^2 + (2(2) + 1)^2 + (2(3) + 1)^2 \\ & + (2(4) + 1)^2 + (2(5) + 1)^2 + (2(6) + 1)^2 \\ & + (2(7) + 1)^2 + (2(8) + 1)^2 + (2(9) + 1)^2 \\ & + (2(10) + 1)^2 + (2(11) + 1)^2 + (2(12) + 1)^2 \\ & + (2(13) + 1)^2 + (2(14) + 1)^2 + (2(15) + 1)^2 \\ & + (2(16) + 1)^2 + (2(17) + 1)^2 + (2(18) + 1)^2 \\ & + (2(19) + 1)^2 + (2(20) + 1)^2 + (2(21) + 1)^2 \end{aligned}$$

$$\begin{aligned} b = & (1)^2 + (2 + 1)^2 + (4 + 1)^2 + (6 + 1)^2 + (8 + 1)^2 + (10 + 1)^2 \\ & + (12 + 1)^2 + (14 + 1)^2 + (16 + 1)^2 + (18 + 1)^2 \\ & + (20 + 1)^2 + (22 + 1)^2 + (24 + 1)^2 + (26 + 1)^2 \\ & + (28 + 1)^2 + (30 + 1)^2 + (32 + 1)^2 + (34 + 1)^2 \\ & + (36 + 1)^2 + (38 + 1)^2 + (40 + 1)^2 + (42 + 1)^2 \end{aligned}$$

$$\begin{aligned} b = & 1 + (3)^2 + (5)^2 + (7)^2 + (9)^2 + (11)^2 + (13)^2 + (15)^2 \\ & + (17)^2 + (19)^2 + (21)^2 + (23)^2 + (25)^2 + (27)^2 \\ & + (29)^2 + (31)^2 + (33)^2 + (35)^2 + (37)^2 + (39)^2 \\ & + (41)^2 + (43)^2 \end{aligned}$$

$$\begin{aligned} b = & 1 + 9 + 25 + 49 + 81 + 121 + 169 + 225 + 289 + 361 + 441 \\ & + 529 + 625 + 729 + 841 + 961 + 1089 + 1225 \\ & + 1369 + 1521 + 1681 + 1849 \end{aligned}$$

$$b = 14190 \text{ blocks}^2$$

Find σ_l .

$$\sigma_l = b - 1$$

$$\sigma_l = 14190 - 1$$

$$\sigma_l = 14189 \text{ blocks}^2$$

Statement:

The area of the total blocks of a beacon pyramid with $A_s = 1 \text{ blocks}^2$ in layer 21 is 1849 blocks^2 . The total blocks needed to build a 22 blocks tall beacon pyramid is 14190 blocks^2 . Excluding the beacon, the total blocks will be 14189 blocks^2 .